

Zwischenpräsentation

Snippet 1:

```
const particles_size = 10;
const resolution = 10;
const Max_Force = 10;
const Min_Force = 0;

let pullSize = 500

let img;
let particles = [];

function preload() {
  img = loadImage('/data/TYP0_PICS/Text_Images4.jpg');
}

function setup() {
  createCanvas(windowWidth, windowHeight)
  setupAudio(true) // if empty, use 'a5.' before audio vars below
  // a5.ease = .075 // customize ease speed
  spawnParticles();
}

function draw() {
  updateAudio()
  background(40);
  //image(img, 0, 0);
  particles.forEach((particle)=>{
    particle.update();
    particle.draw();
  })
}

function spawnParticles(){
  for(let i = 0; i < width; i += resolution){
    for(let j = 0; j < height; j += resolution){
      let x = i/width*img.width
```

```

        let y = j/height*img.height

        const color = img.get(x, y);
        particles.push(new Particle(i + particles_size/2, j +
particles_size/2, color));
    }
}

class Particle{
    constructor(x, y, color){
        this.x = x;
        this.y = y;
        this.color = color;
        this.targetX = x;
        this.targetY = y;
    }
    update(){
        //get vectors for mouse, target and current position
        let mouseVector = createVector(mouseX, mouseY);
        let currentVector = createVector(this.x, this.y)
        let targetVector = createVector(this.targetX, this.targetY);

        //calculate vector from mouse to particle and its magnitude
        (distance)
        let fromMousetoParticle = p5.Vector.sub(currentVector,
mouseVector);
        let distancetoMouse = fromMousetoParticle.mag();

        //calculate vector to target and calculate its magnitude
        let fromParticletoTarget = p5.Vector.sub(targetVector,
currentVector);
        let distancetoTarget = fromParticletoTarget.mag();
        let totalForce = createVector(0, 0);

        //if mouse is within 100 pixels, calculate a repulsive force
        if(distancetoMouse < amp*2){
            let repulsionForce = map(distancetoMouse, 0, amp, Max_Force,
Min_Force)
            fromMousetoParticle.setMag(repulsionForce);
            totalForce.add(fromMousetoParticle);
        }
        //if particle is not a target, calculate an attractive force
        if(distancetoMouse > 0){

```

```

        let attractionForce = map(distancetoTarget, 0, 50, Min_Force,
Max_Force)

        fromParticletoTarget.setMag(attractionForce);
        totalForce.add(fromParticletoTarget);
    }
    //add the forces to the position
    this.x += totalForce.x;
    this.y += totalForce.y;
}
draw(){
    fill(this.color);
    noStroke();
    ellipse(this.x, this.y, particles_size);
}
}

```

![P5L_typo_particles_beat_20260526174915]

(/Users/juliansteiner/Documents/Studium/2BA/Creative Coding/Week
2/images/P5L_typo_particles_beat_20260526174915.png)

Snippet 2:

```

const elementY = 100;
var charset = "0000000000-----0000000000";
let font;

function setup() {
    createCanvas(windowWidth, windowHeight)
    setupAudio(true) // global vars
    // a5.ease = .075 // set easing
}

function draw() {
    background(255, 255, 0)

    var elements = 10;
    fill(0)
    noStroke()

    textFont();
    textAlign(CENTER, CENTER);
    textSize(height / elementY);

    for(let y = 0; y < elementY + 1; y++) {

```

```
for(let x = 0; x < charset.length + 1; x++) {  
  
    let posY = map(y, 0, elementY, 0, height);  
    let magX = map(sin(radians(posY * 0.5 + frameCount*0.5)), -1,  
1, -width * 0.4, width * 0.2);  
    let posX = map(x, 0, charset.length, -magX, magX);  
  
    let selector = x;  
  
    push()  
    translate(width / 2 + posX, posY)  
    text(charset[selector], 0, 0)  
    pop()  
}  
}  
}
```

![P5L_star_01_20260526174843](/Users/juliansteiner/Documents/Studium/2BA/Creative Coding/Week 3/images/P5L_star_01_20260526174843.png)